

Earthworm burrows enhanced rice root growth under a no-tillage rice-rapeseed rotation system

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ABSTRACT

Earthworm burrows are an important class of biopores that facilitate gas and water transport and affect plant root growth. However, in paddy soils that experience periodic flooding during the rice growth season, the effects of earthworm burrows on rice root growth have often been overlooked. In this study, we used X-ray Computed Tomography (CT) to investigate the impact of earthworm (*Metaphire guillelmi*) burrows on rice (*Oryza sativa* L.) root growth. Undisturbed soil cores (0–36 cm depth), collected from rotary tillage (RT) and no-tillage (NT) paddy fields, was used for a greenhouse incubation experiment that simulated a rice-rapeseed rotation system either with (RTE, NTE) or without (RTN, NTN) the addition of earthworms during the rapeseed (*Brassica napus*) season. Earthworm burrows, rice roots and their interaction were determined with CT imaging in the rice season. RT soil had 1.7 times greater soil macroporosity (>146 µm) than the NT soil. After earthworm addition, the volume and surface area of earthworm burrows in the RTE treatment were 2.4 and 1.7 times greater, respectively, than those in the NTE treatment. The newly formed earthworm burrows in the RTE treatment, however, did not significantly affect rice root traits or their utilization of these burrows. On the contrary, the increased earthworm burrows in the NTE treatment significantly enhanced rice root volume, surface area, length, mean diameter, and the utilization of these burrows by roots was 189% greater than in the NTN treatment. Additionally, the increase in earthworm burrows of the NTE treatment notably increased root density in the subsoil. Our results show that the increase of earthworm burrows has an important role in promoting rice root growth under NT, suggesting earthworms are crucial to the success of no-tillage system for the rice paddies.

1. Introduction

Paddy soil occupies 28% of the national arable land and contributes to 43% of the total grain yield in China (Yousaf et al., 2016). Rice-rapeseed rotation is a widely practiced cropping system on the paddy soil in central and south China. The paddy soils are puddled in May before transplanting rice and rotary-tilled in October before the transplant of rapeseed seedlings. No-tillage has gained the attention of farmers in this region because of less labor, energy, and time costs compared to conventional tillage practices (Chen et al., 2012).

With no-tillage, biological activity is an important factor to soil

structure regeneration. Earthworms are recognized as soil ecosystem engineers (Le Bayon et al., 2021) that can alter soil pore structure through burrowing and casting activities. They also have significant effects on enhancing soil aggregation, promoting crop root growth, and stimulating soil microbial activity (Bertrand et al., 2015; Le Bayon et al., 2021). However, tillage has a substantial disruptive impact on earthworm species richness (Chan, 2001; Crittenden et al., 2014) and earthworm burrow structures (Langmaack et al., 2002). The intensity of tillage acts as a decisive factor in driving changes in earthworm populations (Curry, 2004). In comparison to tilled soils, soils under no-till systems have been shown to exhibit greater earthworm abundance

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(van Capelle et al., 2012; Pelosi et al., 2014). This has been attributed to retaining crop residues on the soil surface that create a milder microclimate, higher soil humidity, and greater organic residue inputs (Langmaack et al., 1999). While considerable studies on the impact of tillage on earthworms have been carried out in upland soil, relatively little is known about tillage effects on earthworms in the paddy soils. Recent studies found earthworms can flourish in a no-tillage rice-rapeseed rotation system (Huang et al., 2018; Li et al., 2022). During the rice growing season, when soil experiences continuous flooding, some earthworms can survive by migrating to the field ridges, which are the banks between paddy fields and about 30 cm above the soil surface. Once drained for rapeseed cultivation after rice has been harvested, the earthworm population recovers in the un-tilled soil, benefiting from the fresh input of residues (Li et al., 2022).

The strong vertical connectivity of earthworm burrows in soil (Bottinelli et al., 2017) provides zones with low mechanical resistance for root penetration in compacted soil, thereby forming preferential channels for root growth (Hirth et al., 1997; Springett and Gray, 1997). This holds particular significance in arable land, as soil mechanical resistance has been identified as a primary constraint for crop root exploration (Hoad et al., 2001; Unger and Kaspar, 1994). Additionally, earthworm burrows may offer an appealing environment for root growth, given their provision of elevated oxygen concentrations in the gas phase and heightened nutrient concentrations in the drilosphere (Stewart et al., 1999). Agapit et al. (2018) investigated the effect of earthworms on rice root system morphology and found that earthworms increased the relative length of fine roots while decreasing the relative length of thick roots. Recently, Petzoldt et al. (2025) studied the influence of 2.5-year-old biopores (artificial macropores previously colonized by an anecic earthworm) on spring barley root growth, and found that root length density was significantly higher in the burrow sheath (up to 8 mm from the macropore) than in the bulk soil. These studies suggest that earthworm-formed biopores could actively facilitate root growth. While several studies have found strong impacts of biopores or macropores (aperture, density, morphology, etc.) on root growth (Atkinson et al., 2020; Koch et al., 2021; Xiong et al., 2022), including rice, the work has mainly been limited to laboratory experiments with artificial biopores in repacked soil (Islam et al., 2021). To date, a quantitative investigation into the impact of earthworm burrows on rice root growth in undisturbed paddy soils remain scarce.

For this greenhouse study, we collected undisturbed soil columns from paddy fields with contrasting mechanical rotary tillage or no-tillage cultivation practices. Using the intact columns, we then conducted a greenhouse incubation experiment where earthworms were either added or excluded. The aims were to investigate (1) the effect of earthworms on soil biopore formation, and (2) the influence of earthworm burrows on rice root growth. The research is highly significant to guiding the adoption of no-tillage in rice rotations, where earthworms could potentially have a major impact.

2. Material and methods

2.1. Experimental site and soil

Soil samples were collected from a field experiment in Taipingqiao Village, Nan County, Hunan Province, China (112°24'E, 29°09'N). This region has a humid subtropical monsoon climate, with an annual average temperature and precipitation of 16.6°C and 1238 mm, respectively. The typical cropping system is rice-rapeseed rotation, with rice grown from June to October and rapeseed grown from October to the following May. The soil from the top 0–20 cm layer has a purple calcareous clay texture with 3.7% sand, 44.4% silt, and 51.9% clay (Fluvisol, FAO taxonomy). The soil pH was 7.9, and the contents of soil organic carbon, available N, P and K were 27.3 g kg⁻¹, 240.3 mg kg⁻¹, 11.9 mg kg⁻¹, and 81.2 mg kg⁻¹, respectively. The field experiment was established in 2016, including two tillage treatments: no-tillage (NT)

and rotary tillage (RT). In the NT treatment, both rice and rapeseed were direct seeded manually. In the RT treatment, the paddy soil was puddled before rice transplanting, and was rotary tilled to a depth of 13 cm before rapeseed planting. A detailed description of the field fertilization and management practices can be found in Huang et al. (2018).

2.2. Soil core preparation and experimental design

Six undisturbed soil cores were collected from each of the two treatments (RT and NT) of the field tillage experiment after the rice harvest in October 2018. Undisturbed soil cores were taken at a depth of 0–36 cm using polyvinyl chloride (PVC) tubes (40 cm in length x 15 cm in internal diameter). The PVC tubes were gradually driven into the soil using a hammer, with the surrounding soil was carefully excavated to reduce resistance. The soil cores (~36 cm) were collected, with 4 cm of the PVC tube left above soil surface. Soil cores were wrapped with cling film to prevent water evaporation and were then carefully transported to the laboratory where the first round of X-ray CT imaging was conducted to quantify soil macropore. After CT scanning, the columns were flooded for two days to expel earthworms in the columns, and then were allowed to drain freely to reach a soil water content of 0.30 cm³ cm⁻³. In addition, small PVC cylinders (5 cm in height, 4.8 cm in internal diameter) were used to collect soil samples from the 0–5, 5–10, 10–15, 15–20 and 20–25 cm soil layer in NT and RT treatments, the soil cores were oven-dried to measure soil bulk density, each treatment had three replicates.

Four treatments were established based on the cores from the field experiment: (1) soil cores collected from the RT treatment and no earthworm was added, RTN; (2) soil cores collected from the RT treatment and earthworms were added, RTE; (3) soil cores collected from the NT treatment and no earthworm was added, NTN; (4) soil cores collected from the NT treatment and earthworms were added, NTE. Each treatment had three replicates. Anecic earthworms (*Metaphire guillelmi*) collected from the field experimental site were used. For the RTE and NTE treatments, two active adult earthworms (with clitellum) of about 2 g per individual were added to the surface of each column, reflecting the earthworm density in the field experiment (Li et al., 2022). The bottom cylinder was sealed using a PVC plate, and a small hole was drilled in the side bottom of the cylinder to drain water when needed. To feed the earthworms, 13 g (dry weight) of air-dried rice straw with a maximum length of 3 cm was added at the soil surface. A 10 cm height transparent plastic ring was used as a guard above the column to prevent earthworm escape. About 3 cm thick black insulation cotton was wrapped around the outer wall of the column to mitigate temperature fluctuations. Earthworms were incubated for one week in a side-opened greenhouse with ambient temperature ranging from 10°C to 23°C before the rapeseeds were sown.

2.3. Plant material and growth condition

Five seeds of the rapeseed cultivar Zhongyouza 19 (*Brassica napus*) were sown at 5 mm depth below the soil surface in each column. After the rapeseeds had emerged, one plant was retained in each column. The rapeseed plants were grown for 22 weeks from 8 November, 2018–10 April, 2019 in a side-opened greenhouse. During this period, the ambient temperature ranged from −3°C to 31°C. An EM50 moisture probe (Meter, USA) was used to monitor water content of the topsoil, which was kept at 0.30 cm³ cm⁻³ by adding water.

Following rapeseed harvest, rice seeds of cultivar Shenliangyou 5814 (*Oryza sativa* L.) were germinated on moist filter paper at 30°C for 48 h before being planted at 5 mm depth below soil surface into the columns. The rice plants were grown for 18 weeks from 25 May to 28 September 2019 in a side-opened greenhouse. During this period, the ambient temperature ranged from 15°C to 39°C. The columns were flooded during the rice growing period, water was added once daily to maintain a water table of 2 cm above soil surface. One controlled drainage event (simulating field “dry-down”) was realized via the bottom drainage

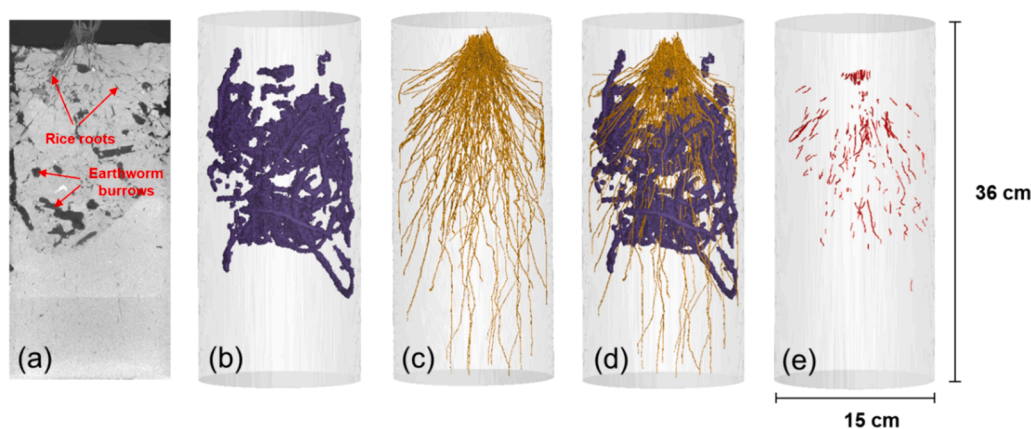


Fig. 1. Extraction of earthworm burrows, roots and their intersections. Example of 2-D X-ray computer tomography grayscale image (a), 3-D earthworm burrow structure (b), 3-D rice roots architecture (c), root-earthworm burrow combinations (d) and the interacted area (e).

holes during the late tillering stage. Rice shoot dry mass, plant height and tiller number were obtained at the end of the experiment. The second round of X-ray CT imaging was conducted to obtain newly formed earthworm burrows and rice root architecture.

2.4. X-ray CT imaging

Soil cores were scanned using a v|tome|x M 300 kV X-ray μ CT scanner (GE Sensing & Inspection Technologies GmbH, Wunstorf, Germany). The first round of scans was performed to detect soil macropores at a resolution of 146 μ m. The second round of scan was performed at a higher resolution of 85.6 μ m with the aim to capture rice roots and newly formed earthworm burrows. For the first round of scans, the voltage and current were 240 kV and 215 μ A, respectively, and the exposure time was 1 s, with a 0.15 mm Cu filter was used to reduce the beam hardening effect. A total of 1400 projection images were collected during the rotation of each sample. For the second round of scans, the scanner was set at 245 kV and 350 μ A, the exposure time was 334 ms, with a 0.15 mm copper filter and an image averaging of 3. A total of 1200 projection images were collected during each scan. Multi-scan was

performed for each set of both round of scans because the height of the soil column (~ 36 cm) was larger than the field of view of the scanner, and the “multi-scan” reconstruction function in Phoenix datos|x software (GE Sensing and Inspection Technologies, Wunstorf, Germany) was used to give a 3D 16-bit greyscale volume. The 3D volume files were exported as 16-bit grayscale slices (tiff format) using VG StudioMAX 3.4 software (Volume Graphics GmbH, Heidelberg, Germany).

2.5. Image analysis and data collection

For pore characteristics, a region of interest (ROI) of 11.7×11.7 cm with a depth of 1–21.9 cm was selected from each column of the NT and RT treatments. The boundaries of the samples were abandoned to avoid disturbance that might have occurred during sampling. A 3D median filter with a radius of 2 voxels was used to suppress noise during image processing. Segmentation of the grayscale images was performed with “Default” thresholding method, a variation on the “IsoData” method where the average of the object and background image are used to compute the threshold (Fang et al., 2019). Pore size distribution was computed using the “Thickness” of BoneJ.

Rice roots, earthworm burrows and their intersected area were segmented using by the VG StudioMax 3.4 software. The earthworm burrows and roots were visually identified on the grayscale images based on their grayscale and morphology (Fig. 1). The earthworm burrows were manually segmented using the “Draw” function (Fig. 1b). Rice roots were semi-automatically segmented using the “Region Growing” tool (Fig. 1c). Earthworm burrows and rice roots in combination were shown in Fig. 1d. The burrows used by rice roots was obtained by intersecting the segmented roots with earthworm burrows using the “interact” algorithm (Fig. 1e).

The morphological characteristics of roots and earthworm burrows were calculated. Root volume, root surface area, burrow volume and burrow surface area were obtained with the VG StudioMax 3.4 software. Root length, tortuosity of root path, root growing angle, burrow tortuosity and burrow connectivity were obtained with the ImageJ software (Version 1.52a, National Institute of Health, <http://imagej.nih.gov/ij/>). Roots and earthworm burrows were first skeletonized using the “skel-ton” plugin, root length was calculated as the length of the skeleton. Root tortuosity and burrow tortuosity were calculated as the ratio of branch length to the Euclidean distance between two end points of the corresponding skeleton. Root growth angle was characterized using the angle (θ) between a line that passed the two end points of the branch and the horizontal plane. A detailed description of the calculation was introduced by Zhang et al. (2018). Mean diameter of root and burrow were computed using the “Thickness” in BoneJ plugin. Root number was obtained using RooTh software (Mairhofer et al., 2017). Burrow number

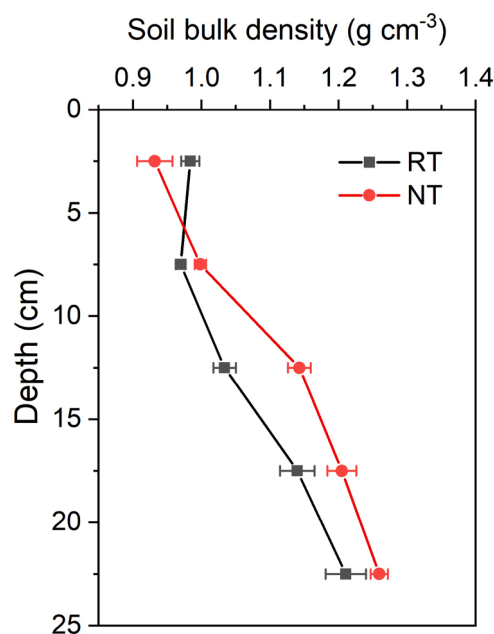


Fig. 2. Soil bulk density from the rotary tillage (RT) and no-tillage (NT) treatment. Error bars show standard error of the mean ($n = 3$).

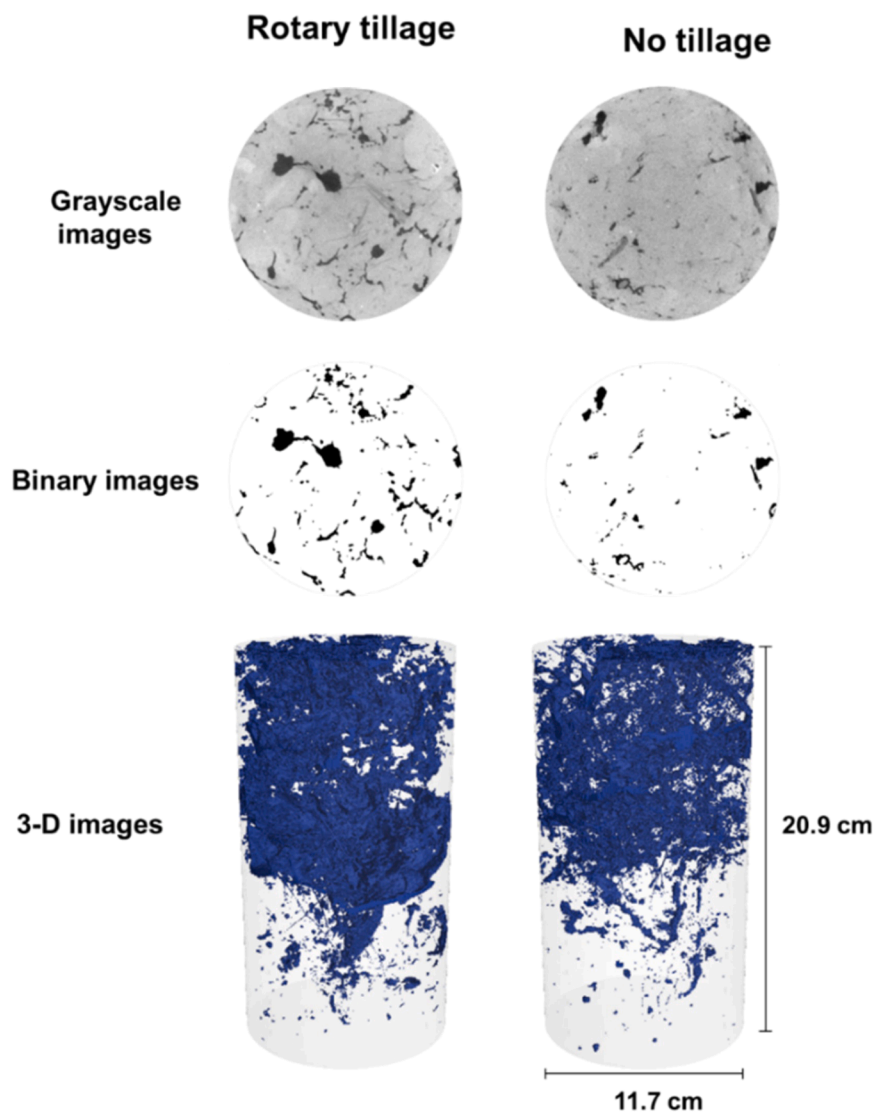


Fig. 3. 2-D grayscale and binary images (diameter: 11.7 cm) and 3-D rendered pore structure (diameter: 11.7 cm, height: 20.9 cm) of the soil from the RT and NT treatments. The pores are shown in dark in grayscale images, black in binary images and blue in 3-D images.

was obtained using the “Particle Analyzer” plugin in ImageJ. The global connectivity of earthworm burrows was calculated as the sum of the squared volume of each individual burrow, divided by the square of the total burrow volume (Fang et al., 2019). The vertical distribution of rice roots and earthworm burrows with soil depth were measured using the “Analyze Particles” plugin in ImageJ.

2.6. Statistical analysis

The normality of residuals and the homogeneity of variances of the data were checked by the Shapiro–Wilk test and the Levene’s test respectively. The *T*-test was used to determine the differences in soil bulk density, and soil macroporosity under the RT soil and NT soil. The impacts of tillage, earthworms and their interactive effects on the earthworm burrows characteristics, and the traits of rice roots and shoots were assessed by using two-way ANOVA at the significance level of $P < 0.05$. The above analyses were completed with the statistical software package IBM SPSS Statistics 20 (IBM, Armonk, NYC, USA).

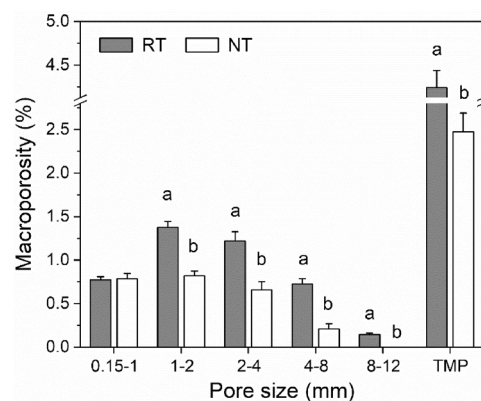


Fig. 4. Macropore size distribution and the total macroporosity (TMP) of the soil from the rotary tillage (RT) and no-tillage (NT) treatment. Lowercase letters indicate significant differences between RT and NT treatments for the same pore size ($P < 0.05$). Error bars show standard error of the mean ($n = 6$).

Table 1

Effects of tillage, earthworm, and their interaction on the earthworm burrows characteristics.

Treatment		Burrow volume (cm ³)	Burrow surface area (cm ²)	Burrow mean diameter (mm)	Burrow connectivity	Burrow tortuosity	Burrow number
Rotary tillage	RTN	38.2(2.02) Ab	720(192) Ab	2.00(0.44) Ab	0.17(0.07) Aa	1.38(0.02) Aa	119(24.0) Ab
	RTE	191(16.5) Aa	2112(55.5) Aa	3.79(0.60) Aa	0.22(0.09) Aa	1.39(0.03) Aa	309(54.5) Aa
No-tillage	NTN	39.5(11.5) Ab	630(202) Ab	2.46(0.30) Aa	0.18(0.05) Aa	1.37(0.01) Aa	97.5(16.5) Ab
	NTE	80.3(9.17) Ba	1265(154) Ba	2.75(0.13) Aa	0.19(0.08) Aa	1.38(0.02) Aa	260(36.5) Aa
P-value	Tillage	**	*	ns	ns	ns	ns
	Earthworm	**	**	*	ns	ns	**
	Tillage × Earthworm	**	ns	ns	ns	ns	ns

Note: RTN: soil cores from the rotary tillage (RT) treatment and no earthworm were added, RTE: soil cores from RT treatment and earthworms were added, NTN: soil cores from the no-tillage (NT) treatment and no earthworm were added, NTE: soil cores from NT treatment and earthworms were added. Uppercase letters indicate significant differences between different tillage soil under the same earthworm treatments, lowercase letters indicate significant differences between different earthworm treatments under the same tillage soil ($P < 0.05$). Numbers in brackets are standard error of the mean ($n = 3$). “ns”, “*” and “**” represent for P -values > 0.05 , < 0.05 and < 0.01 , respectively.

Table 2

Effects of tillage, earthworm, and their interaction on the above-ground biomass, plant height, and tiller number of rice.

Treatment		Above-ground biomass (g)	Plant height (cm)	Tiller number
Rotary tillage	RTN	32.6(1.92) Aa	86.7(1.05) Ba	17.3(0.67) Aa
	RTE	28.7(0.18) Aa	89.9(0.75) Aa	15.3(2.33) Aa
No-tillage	NTN	23.3(0.05) Ba	95.7(1.55) Aa	11.5(1.50) Ba
	NTE	26.7(1.21) Aa	93.0(1.89) Aa	12.7(1.33) Aa
P-value	Tillage	*	**	*
	Earthworm	ns	ns	ns
	Tillage × Earthworm	ns	ns	ns

Note: RTN: soil cores from the rotary tillage (RT) treatment and no earthworm were added, RTE: soil cores from RT treatment and earthworms were added, NTN: soil cores from the no-tillage (NT) treatment and no earthworm were added, NTE: soil cores from NT treatment and earthworms were added. Uppercase letters indicate significant differences between different tillage soil under the same earthworm treatments, lowercase letters indicate significant differences between different earthworm treatments under the same tillage soil ($P < 0.05$). Numbers in brackets are standard error of the mean ($n = 3$). “ns”, “*” and “**” represent for P -values > 0.05 , < 0.05 and < 0.01 , respectively.

3. Results

3.1. NT soil has a denser structure than the RT soil

The NT soil had a greater soil bulk density at 5–25 cm soil layers than the RT soil (Fig. 2). An example comparison of the 2-D grayscale image, binary image, and 3-D rendered pore structure of the soil from the NT and RT treatments are shown in Fig. 3. Soil macropores were mainly distributed at 0–10 cm soil depth. There were more macropores observed under the RT treatment compared to the NT treatment. Soil macroporosity and the macropore size distribution are shown in Fig. 4. Total macroporosity ($> 146 \mu\text{m}$) of the RT soil was 1.7 times greater than that of the NT soil ($P < 0.05$). No significant differences in the porosity of the 0.15–1 mm size were observed between the RT and NT treatments. However, the RT treatment had a much greater porosity in the pores of size 1–2 mm, 2–4 mm, 4–8 mm, and > 8 mm than in the NT treatment ($P < 0.05$).

3.2. Less newly formed earthworm burrows were in the NT soil than in RT soil

Soil cores with added earthworms (RTE and NTE) had more burrows compared to the soil cores without earthworms (RTN and NTN), indicating the added earthworms created new burrows. Compared to the

corresponding no earthworm treatments, the volume of burrows increased by 400% for RTE ($P < 0.05$) and 103% for NTE ($P < 0.05$), and the number of burrows increased by 160% for RTE ($P < 0.05$) and 167% for NTE ($P < 0.05$) (Table 1). There were no significant differences in the volume, surface area and mean diameter of earthworm burrows between the treatments without earthworms (RTN and NTN) (Table 1). However, after the addition of earthworms, more earthworm burrows were observed in the RTE treatment than in the NTE treatment (Fig. 6b and d). The volume and surface area of earthworm burrows in the RTE treatment were 2.4 and 1.7 times greater, respectively, than those in the NTE treatment ($P < 0.05$) (Table 1). Tillage, earthworms, and their interactions had no effect on burrow connectivity and burrow tortuosity (Table 1).

The distribution of the earthworm burrow density with soil depth is shown in Fig. 6 (b and d). The RTE treatment had much greater earthworm burrow density at 5–20 cm depth compared to the RTN treatment. The NTE treatment showed greater earthworm burrow density only at 5–10 cm depth than the NTN treatment.

3.3. Earthworm burrows in no-till soil influenced the growth of rice roots

The above-ground biomass, plant height and tiller number of the rice plants were significantly affected by tillage practice, but not by earthworms nor earthworm × tillage interaction (Table 2). Post-hoc analysis revealed that above-ground rice biomass and tiller number in the RTN treatment were significantly increased by 37.8% and 50.7% than those in the NTN treatment ($P < 0.05$), respectively, while the plant height in the RTN treatment was reduced by 9.4% compared to both the NT treatments ($P < 0.05$). No significant differences in above-ground biomass, plant height and tiller number were observed between the RTE and NTE treatments.

The three-dimensional rice root architecture is shown in Fig. 5 and the traits of rice roots are shown in Table 3. It was found from visual inspection of the X-ray CT images that roots grew deeper in the NTE treatment than in the NTN treatment (Fig. 5). Two-way ANOVA revealed that tillage, earthworms, and their interactions affected rice root volume and surface area (Table 3). The number of roots and root length were also affected by tillage and tillage × earthworm interactions, but not by earthworms alone. Root diameter was only significantly affected by earthworms. In the rotary tillage practice, no significant differences in the root traits were found between the RTE and RTN treatments. However, in the no-tillage soil, root volume, surface area, length, number, and mean diameter of the NTE treatment were significantly greater by 36.0%, 34.5%, 37.2%, 29.8%, and 10.3% than the NTN treatment ($P < 0.05$). Comparing the four treatments, the NTN treatment had the smallest root volume, surface area, length, numbers and mean diameter, while the differences among the other three treatments were not significantly different. The vertical distribution of rice roots with soil depth (Fig. 6a and c) showed that the root density of each soil layer was

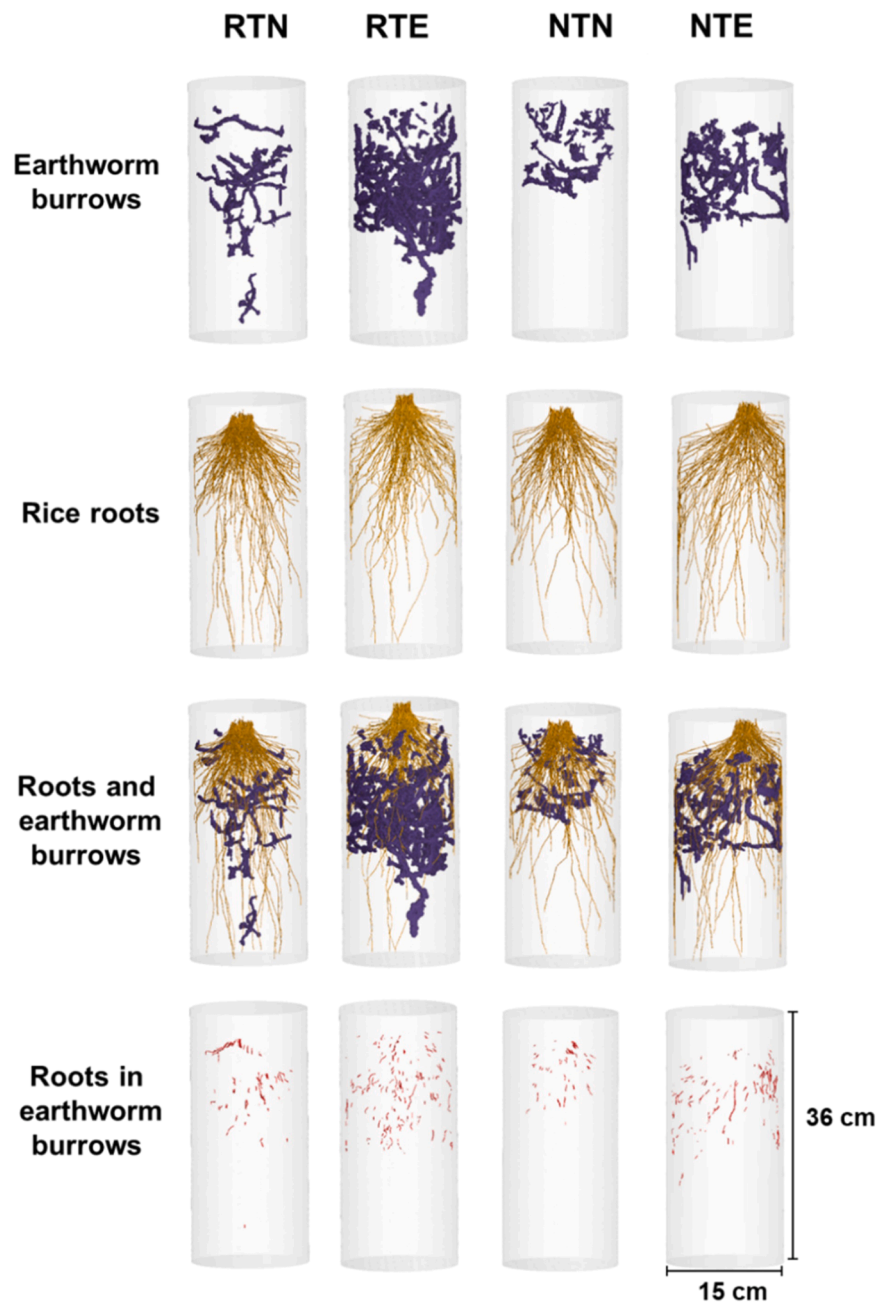


Fig. 5. Representative 3-D root architecture and earthworm burrow of the different treatments. Purple color represents earthworm burrows, yellow color represents rice roots, and red color represents the intersecting region of the rice roots and earthworm burrows. The volume interacted by roots and earthworm burrows under different treatments. RTN: soil cores from the rotary tillage (RT) treatment and no earthworm were added, RTE: soil cores from RT treatment and earthworms were added, NTN: soil cores from the no-tillage (NT) treatment and no earthworm were added, NTE: soil cores from NT treatment and earthworms were added.

affected by earthworms for NT but not RT. No significant differences in root angle and root tortuosity were observed among the four treatments (Table 3).

The interaction between roots and earthworm burrows is visualized in Fig. 5 and quantitative results are shown in Fig. 7. When no earthworm was added, the intersection volume between roots and burrows was the same for RT and NT. Following the addition of earthworms, no significant difference emerged between the RTN and RTE treatments, but the intersection in NTE was 189% greater than in the NTN treatment ($P < 0.05$). The NTE treatment had the greatest volume of root-earthworm burrow intersection among the four treatments. However, there was no significant difference between the two earthworm treatments (RTE and NTE).

4. Discussion

The change from rotary tillage to no-tillage explored in our study resulted in distinct soil structures in a system where flooded rice forms part of the rotation. Mechanical loosening by rotary tillage had the expected result of lower soil bulk density and higher macroporosity compared to no-tillage soils (Figs. 3 and 4), in agreement with results of previous studies of upland soils (Sasal et al., 2006; Mangalassery et al., 2014; Pohlitz et al., 2018). Pores smaller than 0.146 mm were not considered in this study due to the imaging resolution limit (0.146 mm) for the large size of the studied soil columns (15 cm in diameter and 40 cm in height). Our study showed that the rotary-tillage soil had more > 2 mm macropores, which may be due to more cracks in rotary-tillage

Table 3
Effects of tillage, earthworm, and their interaction on the rice root traits.

Treatment		Root volume (cm ³)	Root surface area (cm ²)	Root length (cm)	Root mean diameter (mm)	Root number	Root angle	Root tortuosity
Rotary tillage	RTN	16.6(0.28) Aa	790(19.6) Aa	31.6(0.11) Aa	0.70(0.01) Aa	306(15.0) Aa	55.4(1.49) Aa	1.23(0.11) Aa
	RTE	16.6(0.50) Aa	769(19.0) Aa	29.2(1.63) Aa	0.74(0.01) Aa	278(6.50) Aa	55.3(1.16) Aa	1.22(0.11) Aa
No-tillage	NTN	11.4(0.51) Bb	565(27.1) Bb	22.3(1.41) Bb	0.68(0.02) Ab	205(13.5) Bb	57.8(1.75) Aa	1.22(0.02) Aa
	NTE	15.5(0.38) Aa	760(10.5) Aa	30.6(1.14) Aa	0.75(0.01) Aa	266(4.50) Aa	56.1(0.11) Aa	1.22(0.01) Aa
P-value	Tillage	**	**	*	ns	**	ns	ns
	Earthworm	**	*	ns	**	ns	ns	ns
	Tillage	**	**	*	ns	ns	ns	ns
	× Earthworm							

Note: RTN: soil cores from the rotary tillage (RT) treatment and no earthworm were added, RTE: soil cores from RT treatment and earthworms were added, NTN: soil cores from the no-tillage (NT) treatment and no earthworm were added, NTE: soil cores from NT treatment and earthworms were added. Uppercase letters indicate significant differences between different tillage soil under the same earthworm treatments, lowercase letters indicate significant differences between different earthworm treatments under the same tillage soil ($P < 0.05$). Numbers in brackets are standard error of the mean ($n = 3$). “ns”, “*” and “**” represent for P -values > 0.05 , < 0.05 and < 0.01 , respectively.

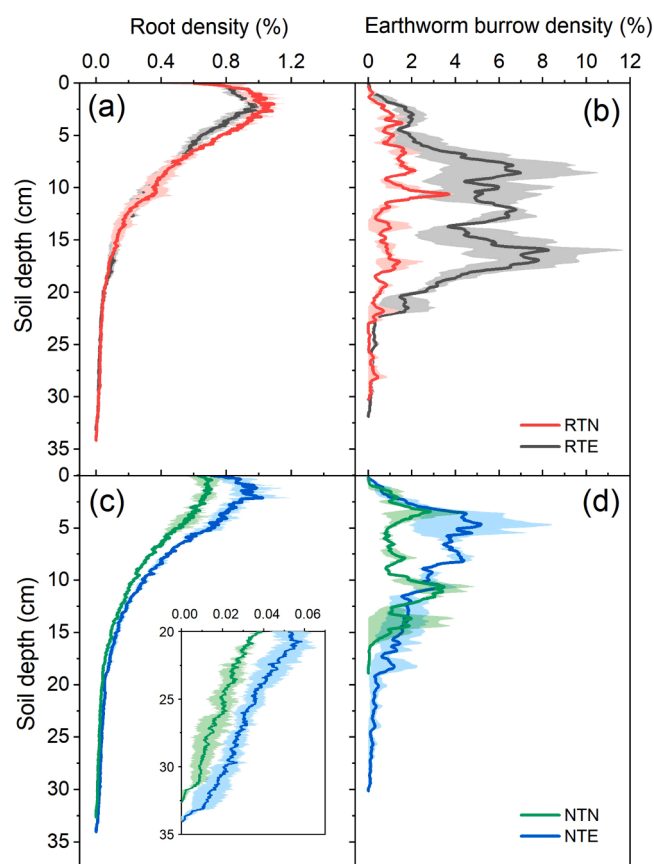


Fig. 6. Distribution of root density (a and c) and earthworm burrow density (b and d) under different treatments to a depth of 35 cm. The volume interacted by roots and earthworm burrows under different treatments. RTN: soil cores from the rotary tillage (RT) treatment and no earthworm were added, RTE: soil cores from RT treatment and earthworms were added, NTN: soil cores from the no-tillage (NT) treatment and no earthworm were added, NTE: soil cores from NT treatment and earthworms were added. The inset in (c) shows an upscaled view of the distribution of root density in 20–35 cm. The shaded areas are the standard error of the mean ($n = 3$).

soil. Tillage in paddy fields (such as puddling before the rice season and rotary tillage before the rapeseed season) leads to the formation of cracks. The wetting and drying cycles during the rice season also

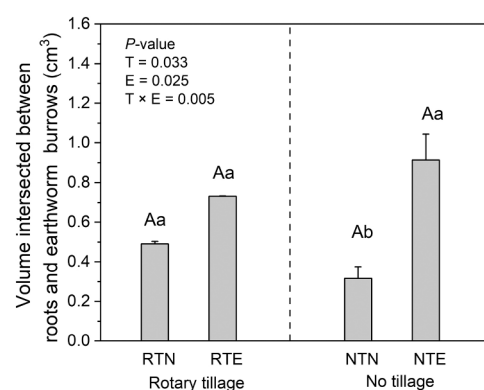


Fig. 7. The volume interacted by roots and earthworm burrows under different treatments. RTN: soil cores from the rotary tillage (RT) treatment and no earthworm were added, RTE: soil cores from RT treatment and earthworms were added, NTN: soil cores from the no-tillage (NT) treatment and no earthworm were added, NTE: soil cores from NT treatment and earthworms were added. T: tillage, E: earthworm. Error bars show standard error of the mean ($n = 3$). Uppercase letters indicate significant differences between different tillage soil under the same earthworm treatments, lowercase letters indicate significant differences between different earthworm treatments under the same tillage soil ($P < 0.05$).

promotes crack formation in puddling soil (Fang et al., 2018).

For no-till soil, our study found that the addition of earthworms significantly increased the volume, surface area, length, and number of rice roots compared to the treatment without earthworms (Table 1). Agapit et al. (2018) also observed that earthworms increased the biomass of rice roots. In a study by Petzoldt and Kautz (2021), roots of faba bean were observed to grow in earthworm burrows. Petzoldt et al. (2025) further demonstrated that the burrow sheath, enriched in total carbon and nitrogen, significantly enhanced the root length density of spring barley compared to the bulk soil. The stimulation of rice root growth by earthworms could be explained by several potential reasons. As dense soil under no-tillage systems (Fig. 2) might limit root development (Lipiec et al., 2012), the biopores formed by the earthworms could provide channels with lower mechanical resistance and higher oxygen availability. This explanation was supported by significantly promoted root elongation and branching in macropores (Colombi et al., 2017; Fang et al., 2018; Liu et al., 2024). In addition, as soil mechanical resistance increases, earthworms expend more energy to move through the soil, leading to greater cast production (Arrazola-Vasquez et al.,

2022). These casts, which were rich in minerals, polysaccharides, and organic matter, could support root growth (van Groenigen et al., 2019). Moreover, the walls of earthworm burrows have been found to be enriched with essential nutrients such as nitrogen, phosphorus, potassium, and calcium (Kautz, 2014).

From the visual X-ray CT images (Fig. 5) and quantitative root analysis (Fig. 6) and an important observation in our study was that more roots grew deeper in the no-tillage soil when earthworms were added (Fig. 5). This trend was not observed in the rotary-tillage soil (Fig. 6a), despite a smaller increase in earthworm burrows under the no-tillage (NTE vs. NTN) compared to the rotary tillage (RTE vs. RTN) (Table 1). This suggests that the increase in earthworm burrows of the no-till soil promoted root growth, while for the rotary-tilled soil, root development was less affected by earthworm burrows. This could be attributed to the higher soil mechanical resistance encountered by roots under no-tillage compared to rotary tillage, leading to greater utilization of earthworm burrows by the roots. Moreover, the inherently high porosity of rotary-tilled soil (Fig. 4) likely diminished the positive effects of earthworm burrows. Atkinson et al. (2020) reported that in loose soil with a bulk density of 1.2 g cm^{-3} , maize roots had a colonization rate of only 12.5% in artificial macropores, by 68.8% in compacted soil with a bulk density of 1.6 g cm^{-3} . Similarly, Xiong et al. (2022) found that in non-compacted soil, most roots merely crossed over artificial macropores they introduced in their controlled study. In addition, Phalempin et al. (2022) found that roots showed no preference for biopores formed from previous roots in well-aerated soil. Collectively, these studies suggest that roots utilize macropores or biopores for growth in compacted soil, a finding that aligns with the results of this study.

More importantly, we found that the increase in earthworm burrows significantly enhanced root utilization of biopores (created by earthworms) in the no-tillage soil (NTE vs. NTN), whereas no such effect was observed under the rotary tillage. It is important to note that soil compaction, which is a key limiting factor to root growth, is often associated with continuous no-tillage practices. However, in the long-term, by preserving soil structure and encouraging the development of soil ecological diversity, no-tillage practices can allow earthworms to thrive, and the stability of their burrows tends to increase over time (Pelosi et al., 2017; Huang et al., 2018). Our findings have identified that earthworms played a crucial role in overcoming the limitations of long-term no-tillage in rice paddies, ensuring better root growth, particularly at depth. Therefore, we suggest that integrating earthworms with no-tillage practice could offer a promising alternative to traditional tillage practices.

5. Conclusion

Rice paddy rotations with upland crops are moving towards no-tillage cultivation systems. However, negative aspect of this practice is that soils can become denser compacted to traditional tillage methods. Our study has found that in the potentially dense soil that results, earthworms have a vital role in promoting root growth, particularly to deeper into the soil profile. This deeper rooting may also confer additional benefits if drought stress conditions arise. Despite our finding that earthworm activity and therefore burrows were more pronounced under rotary tillage, there was little impact on root growth. On the contrary, the increase in earthworm burrows in the no-tillage soil significantly enhanced root volume, surface area, length, mean diameter, and the utilization of these burrows by roots. Additionally, the increase in earthworm burrows under no-tillage also notably increased the root density in the subsoil and induced a directional effect on root growth. This study contributes to a deeper understanding of the relationship between soil biota, plant root systems, and soil structure, providing unique results for paddy soil under no-tillage cultivation practices.

CRediT authorship contribution statement

Lingling Liu: Writing – review & editing, Writing – original draft. **Hu Zhou:** Writing – review & editing, Supervision. **Paul D. Hallett:** Writing – review & editing. **Craig J. Sturrock:** Writing – review & editing. **Weida Gao:** Writing – review & editing. **Huan Fang:** Methodology. **Chao Li:** Investigation.

Declaration of Competing Interest

The authors declare that there is no conflict of interest regarding the publication of this article.

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Data availability

Data will be made available on request.

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